



Northern Beaches Secondary College

Manly Selective Campus

2012 HSC Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators and templates may be used.
- All necessary working should be shown in every question.
- Multiple choice questions are to be completed on the special answer page.

Total marks – 70

- Attempt Questions 1-14.

Multiple choice questions – answer on the special answer sheet provided.

1. A class consists of 10 girls and 12 boys. How many ways are there of selecting a committee of 3 girls and 2 boys?

- (A) 7920
- (B) 26334
- (C) 95040
- (D) 3160080

2. Which of the following is an expression for $\int \frac{x}{\sqrt{x^2 - 2}} dx$.

Use the substitution $u = x^2 - 2$.

- (A) $\sqrt{x^2 - 2} + c$.
- (B) $(\sqrt{x^2 - 2})^3 + c$.
- (C) $2\sqrt{x^2 - 2} + c$.
- (D) $2(\sqrt{x^2 - 2})^3 + c$.

3. Seven people are seated around a round table. How many arrangements are possible if two people refuse to sit beside each other?

- (A) 24
- (B) 120
- (C) 240
- (D) 480

4. The expression $\tan\left(\frac{\pi}{4} + x\right) =$ can be also expressed as

- (A) $\frac{\cos x + \sin x}{\cos x - \sin x}$
 (B) $\frac{\cos x - \sin x}{\cos x + \sin x}$
 (C) $\frac{\sec^2 x}{1 - \tan^2 x}$
 (D) $\frac{\sin x + \cos x}{\sin x - \cos x}$

5. Determine the acute angle between the lines $2x + y = 4$ and $x - 3y = 2$ to the nearest degree.

- (A) 82°
 (B) 45°
 (C) 18°
 (D) 72°

6. What is the exact value of the definite integral $\int_{\frac{2}{\sqrt{3}}}^{2\sqrt{3}} \frac{dx}{x^2 + 4}$?

- (A) $\frac{\pi}{12}$
 (B) $\frac{\pi}{6}$
 (C) $\frac{\pi}{3}$
 (D) $\frac{\pi}{2}$

7. Which of the following is true for the equation of motion $y = 3 \sin \frac{2}{3}t$?

- (A) period = $\frac{2}{3}$.
 (B) period = $\frac{4}{3}$.
 (C) period = 3.
 (D) the period is none of the above values.

8. $\lim_{x \rightarrow 0} 3x \operatorname{cosec} \frac{x}{2} =$

- (A) $\frac{2}{3}$.
 (B) $\frac{1}{6}$.
 (C) $\frac{3}{2}$.
 (D) 6.

9. The length of each side of a cube is increasing at a rate of 1 mm/min. At what rate is the volume increasing when the side edge is 10 cm?

- (A) $6 \text{ cm}^3/\text{min}$
 (B) $12 \text{ cm}^3/\text{min}$.
 (C) $30 \text{ cm}^3/\text{min}$
 (D) none of the above values.

10. Which of the following expressions is correct?

(A) $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1-x^2}}$

(B) $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$

(C) $\tan^{-1} x = \cos^{-1} \frac{x}{\sqrt{1-x^2}}$

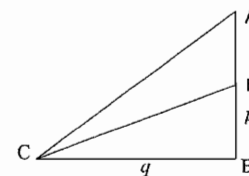
(D) $\tan^{-1} x = \cos^{-1} \frac{x}{\sqrt{1+x^2}}$

Free response questions – answer each question in a separate Booklet

Question 11 (Start a new Booklet)

15

(a) In $\triangle ABC$, $\hat{A}BC = 90^\circ$ and CD bisects $\hat{A}CB$. $BD = p$ and $BC = q$.



Find an expression for $\sin \hat{A}CB$ in terms of p and q . (2)

(b) What is the monic polynomial $Q(x)$ whose roots are (2)

$$2 + \sqrt{3}, 2 - \sqrt{3} \text{ and } 2$$

Express your answer in expanded form.

(c) Using Newton's method with $x = 3$, find a better approximation for the value of x in the equation $\tan x = \log_e x$. (2)

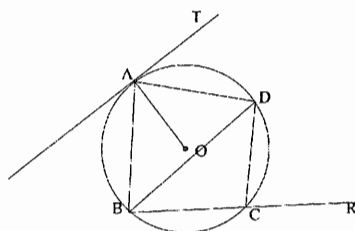
Question 11 (continued)

- (d) Points A, B, C and D lie on a circle with centre O.

The line TA is a tangent to the circle at A and BC is produced to R.

The interval OA bisects $\hat{BAD}$ and $BC = CD$.

Let the size of $\hat{DBC} = \alpha$.



- (i) Show $\hat{OAD} = \alpha$. (2)

- (ii) Prove that $\hat{ABC}$ is a right angle. (2)

- (e) (i) By considering the sum of an arithmetic series, show that (1)

$$(1 + 2 + 3 + \dots + n)^2 = \frac{1}{4}n^2(n+1)^2.$$

- (ii) Hence use the technique of mathematical induction to prove that (4)

$$1^3 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + 3 + \dots + n)^2 \quad n \geq 1$$

Question 12 (Start a new Booklet)

15

- (a) $T(2t, t^2)$ is point on the parabola $x^2 = 4y$ with focus F . P is the point which divides FT internally in the ratio 1:2. Show that the locus of P is a parabola. (4)

- (b) Determine $\int \sin^2\left(\frac{x}{2}\right) dx$ (2)

- (c) A cup of hot coffee at temperature $T^\circ\text{C}$ loses heat when placed in a cooler environment. It cools according to the law

$$\frac{dT}{dt} = k(T - T_0)$$

where t is time elapsed in minutes and T_0 is the temperature of the environment in degrees Celsius.

- (i) Show that the equation $T = T_0 + Ae^{kt}$ is a solution for the differential equation. (1)

- (ii) A cup of coffee at 100°C is placed in an environment at -20°C for 3 minutes during which time it cools to 70°C . Find the exact value of k . (1)

- (ii) The same cup of coffee, at 70°C , is then placed in an environment at 20°C . Assuming k remains the same, find the temperature of the coffee after a further 15 minutes. (2)

- (d) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -2e^{-x}$ where x metres is the displacement from the origin. Initially the particle is at the origin with a velocity of 2 ms^{-1} .

Prove $v = 2e^{-\frac{x}{2}}$. (2)

- (e) Determine the coefficient of x^5 in the expansion of $\left(2x^2 + \frac{1}{x}\right)^7$. (3)

Question 13 (Start a new Booklet)

15

- (a) The velocity of a particle moving along the x axis is given by

$$v^2 = -7 + 8x - x^2$$

- (i) Find an expression for the acceleration of the particle. (2)

- (ii) Explain why the motion of the particle is simple harmonic. (1)

- (iii) Identify the amplitude, period and centre of motion. (2)

- (iii) Find the maximum speed. (1)

- (b) The function $f(x)$ is defined as $f(x) = \operatorname{cosec} x$ $0 \leq x \leq \frac{\pi}{2}$.

- (i) State the range of $f(x)$. (1)

- (ii) Find $f^{-1}(x)$. (2)

- (iii) Hence find $\frac{d}{dx}(f^{-1}(x))$ (2)

- (c) Draw a fully labelled sketch of $y = 4\cos^{-1}(2 - 3x)$. (2)

- (d) Solve the following inequality for: $\frac{2}{x} \geq x - 1$. (2)

Question 14 (Start a new Booklet)

15

- (a) A particle is moving in a straight line. At time t seconds it has displacement x metres to the right of a fixed point O on the line and velocity $v \text{ ms}^{-1}$ given by $v = \sin x \cos x$.

The particle starts at $\frac{\pi}{4}$ metres to the right of O .

- (i) Show that $\frac{d}{dx} \ln(\tan x) = \frac{1}{\sin x \cos x}$ (1)

- (ii) Hence show that the displacement of the particle is given by $x = \tan^{-1}(e^t)$. (2)

- (b) A rescue plane is travelling at an altitude of 120 metres above sea level and a constant speed of 216 km/h towards a stranded sailor. A canister containing a life raft is dropped from the plane to the sailor.

- (i) How long will it take for the canister to hit the water? (Take $g = 10 \text{ ms}^{-2}$) (4)

A current is causing the sailor to drift at a speed of 3.6 kmh^{-1} in the same direction as the plane is travelling. The canister is dropped when the horizontal distance from the plane to the sailor is D metres.

- (ii) What values can D take if canister lands at most 50 metres away from the sailor? (2)

- (c) Find the greatest coefficient in the expansion $(2 + 5x)^{11}$. (2)

- (d) By considering the expansion of $\left(2 + \frac{x}{2}\right)^n$ prove that (4)

$$\left(\frac{5}{2}\right)^n - 1 = \frac{1}{2^2} \binom{n}{1} + \frac{1}{2^4} \binom{n}{2} + \dots + \frac{1}{2^{2n}} \binom{n}{n}$$

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Part A:

Question	Answer	Working
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Question 11

(a)	$\tan \hat{DCB} = \frac{p}{q}$ and $\sin \hat{DCB} = \frac{p}{\sqrt{p^2 + q^2}}$ But CD bisects $\hat{ACB}$ $\therefore \hat{ACB} = 2 \times \hat{DCB} = 2 \times \alpha$ (say) $\therefore \sin 2\alpha = 2 \times \frac{p}{\sqrt{p^2 + q^2}} \times \frac{q}{\sqrt{p^2 + q^2}}$ $= \frac{2pq}{p^2 + q^2}$	2 marks – correct expression. 1 mark – recognition of need for a double angle.
(b)	$\sum \alpha = 6$ $\sum \alpha \beta = (2 + \sqrt{3})(2 - \sqrt{3}) + 2(2 + \sqrt{3}) + 2(2 - \sqrt{3}) = 9$ $\alpha \beta \gamma = (2 + \sqrt{3}) \times 2(2 - \sqrt{3}) = 1 \times 2 \times 1 = 2$ monic polynomial is $x^3 - 6x^2 + 9x - 2 = 0$	2 marks – correct polynomial 1 mark – correct approach using roots and at least one component value correct.
(c)	let $y = \tan x - \log_e x$ $\frac{dy}{dx} = \sec^2 x - \frac{1}{x}$ At $x = 3$ $y = \tan 3 - \ln 3$ $\frac{dy}{dx} = \sec^2 3 - \frac{1}{3}$ REMEMBER radians $y_2 = 3 - \frac{\tan 3 - \ln 3}{\sec^2 3 - \frac{1}{3}}$ $= 4.81$	2 marks – correct approximation. 1 marks – correct derivative and substitutions.
(d)(i)	$BC = CD$ $\therefore \triangle BDC$ is isosceles and $\hat{DBC} = \hat{BCD} = \alpha$ $\hat{RCD} = 2\alpha$ (exterior angle of triangle BCD equals the sum of interior opposite angles). $\hat{DCR} = \hat{BAD} = 2\alpha$ (external angle of a cyclic quadrilateral equals the interior opposite angle). But OA bisects $\hat{BAD}$ $\therefore \hat{OAD} = \alpha$	2 marks – correct proof with justification. 1 mark – correct proof with lack of sufficient justification OR appropriate identification of $\angle DCR$.

(d) (ii)	As $\hat{OAD} = \alpha$ $\hat{TAD} = 90^\circ - \alpha$ $\therefore \hat{ABD} = 90^\circ - \alpha$ (angle between a tangent AT and a cord AD = angle in alternate segment). But $\hat{DBC} = \alpha$ $\therefore \hat{ABC} = 90^\circ$	2 marks – correct proof 1 mark – correct use of tangent theorem.
(e) (i)	arithmetic series with $t_1 = 1$ and $d = 1$ $S_n = \frac{n}{2}(2 + (n-1)) = \frac{n}{2}(n+1)$ $\therefore (1 + 2 + \dots + n)^2 = \frac{n^2}{4}(n+1)^2$	1 mark – correct proof.
(ii)	Try $n = 1$ $LHS = 1^3 = 1$ $RHS = 1^2 = 1 \therefore LHS = RHS$ Assume true for $n = k$ $\therefore (1^3 + 2^3 + 3^3 + \dots + k^3) = (1 + 2 + 3 + \dots + k)^2$ Try $n = k + 1$ SO RTP $1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = (1 + 2 + \dots + k + k + 1)^2$ $= \frac{1}{4}(k+1)^2(k+2)^2$ $LHS = (1 + 2 + 3 + \dots + k)^2 + (k+1)^3$ $= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$ $= \frac{1}{4}(k+1)^2(k^2 + 4k + 4) = \frac{1}{4}(k+1)^2(k+2)^2$ $= RHS$ Hence proved by technique of mathematical induction $n \geq 1$	4 marks – correct demonstration fully set out. 3 marks – correct substitution for assumption 2 marks – correct RTP statement with additional statement included appropriately 1 mark – correct statement for $n = k$.

Question 12

(a)	Coordinates of P $P = \left(\frac{2 \times 0 + 1 \times 2t}{1+2}, \frac{2 \times 1 + 1 \times t^2}{3} \right) = \left(\frac{2t}{3}, \frac{2+t^2}{3} \right)$ Locus of P $x = \frac{2t}{3} \quad \frac{3x}{2} = t$ $y = \frac{1}{3}(2+t^2)$ $= \frac{1}{3}\left(2 + \frac{9x^2}{4}\right)$ $12y = 9x^2 + 8$ $12y - 8 = 9x^2$ $x^2 = \frac{4}{3}\left(y - \frac{2}{3}\right)$	
(b)	$\int \sin^2\left(\frac{x}{2}\right) dx$ $= \int \frac{1}{2} - \frac{1}{2} \cos x \, dx$ $= \frac{x}{2} - \frac{1}{2} \sin x + C$	
(c) - i	$T = T_0 + Ae^{kt}$ $\therefore T - T_0 = Ae^{kt}$ $\frac{dT}{dt} = kAe^{kt}$ $= k(T - T_0)$	

c- (ii)	$t = 0 \quad T_0 = -20 \quad T = 100$ $T = T_0 + Ae^{kt}$ $100 = -20 + Ae^0$ $A = 120$ $T = -20 + 120e^{3k}$ $70 = -20 + 120e^{3k}$ $\ln\left(\frac{3}{4}\right) = 3k$ $k = \frac{1}{3} \ln\left(\frac{3}{4}\right)$	
c-(iii)	$t = 0 \quad T_0 = 20 \quad T = 70$ $T = T_0 + Ae^{kt}$ $100 = 70 + Ae^0$ $A = 30$ $T = 20 + 30e^{15k}$ $= 20 + 120e^{15 \times \ln\left(\frac{3}{4}\right)}$ $= 21.6^\circ \text{C}$	

(d)	$x = -2e^{-x}$ $\frac{d\left(\frac{1}{2}v^2\right)}{dx} = -2e^{-x}$ $\frac{1}{2}v^2 = 2e^{-x} + \frac{C}{2}$ $t = 0, x = 0, v = 2$ $4 = 4 + C$ $C = 0$ $v^2 = 4e^{-x}$ $v = \pm\sqrt{4e^{-x}}$ $= \pm 2e^{-\frac{x}{2}}$ $v = 2e^{-\frac{x}{2}}$ Velocity is initially positive and acceleration approaches 0 with increasing distance therefore velocity remains positive.	
(e)	$\left(2x^2 + \frac{1}{x}\right)^7$ $\therefore$ General Term is ${}^7C_k (2x^2)^{7-k} \cdot \left(\frac{1}{x}\right)^k$ ${}^7C_k (2^{7-k})(x^2)^{7-k} \cdot (x)^{-k} = Ax^5$ ${}^7C_k (2^{7-k})x^{14-3k} = Ax^5$ $\therefore 14 - 3k = 5$ $k = 3$ $\therefore A = {}^7C_3 \times 2^4 = 336$	

Question 13

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Question 14

Question 15

Question 16
